ASSIGNMENT 2: VIDEO ESSAY SCRIPT

THE HUMAN-AI OPERATING MODEL FOR KMG

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Group:	Group DPA-2602 | Business Administration (BA2026)
Target Companies:	Company A: Shell Company B: KazMunayGas (KMG)
Script Word Count:	415

SECTION 1: Opening position

Visual Cue: Title Slide displaying Student Name, Group DPA-2602, and KMG & Shell Corporate Logos.

Hello, my name is Ayaulym Kenzhebay from group DPA-2602. For KazMunayGas, artificial intelligence must be deployed as an operational driver for predictive maintenance and industrial safety, but primary managerial decision-making must remain strictly human due to KMG's sovereign social mandate and its core role as Kazakhstan’s national energy anchor.

SECTION 2: THE PROCESS & BOTTLENECK

Visual Cue: Process Map from Week 5 displaying Atyrau Refinery workflow with highlighted Bottleneck.

The main operational factor for us is the maintenance process of oil refining equipment at KMG’s domestic oil refineries, including the plants in Atyrau and Pavlodar. The main operational obstacle is unplanned refinery downtime caused by the wear and tear of outdated

equipment, which limits fuel supplies to the country and increases maintenance costs. To eliminate this limitation, our key performance indicator (KPI) is to reduce unplanned downtime by 15% and cut maintenance costs by 10% by the end of 2027 using predictive analytics.

SECTION 3: TASK ALLOCATION MATRIX & OPERATIONAL CONTROLS

Visual Cue: Human-AI Task Allocation Matrix from Week 4 showing 4 allocation categories and review points.

Using our Week 4 Task Allocation Matrix, we distribute operational responsibilities across four distinct categories:

First, Human Only: Strategic workforce restructuring and regional employment decisions must remain strictly human, passing the accountability test due to KMG's labor obligations in the Mangystau region.

Second, Human + AI: The allocation of capital expenditures and modeling of the decarbonization process, such as the project to create a hybrid power plant in Zhanaozen with a capacity of 247 megawatts, are carried out under human supervision and are supplemented by scenario analysis using artificial intelligence to reduce financial risks.

Third, AI Executes with Human Review: Predictive failure monitoring and the ABAI system's waterflooding module across 1,660 injection cells. Here, the Human Review Point requires a certified chief engineer to verify AI anomaly flags before initiating any physical equipment shutdown.

Fourth, Automated: Real-time vibration and temperature telemetry collection from refinery sensors. If sensor networks or AI algorithms fail, the Manual Fallback mandates that plant

operators immediately revert to manual hourly logbooks and standard physical safety protocols.

SECTION 4: Comparative transfer analysis

Visual Cue: Transfer Judgment Matrix comparing Shell and KMG (Adopt, Adapt, Pilot, Delay, Reject).

Comparing KMG with Shell plc provides essential comparative evidence. We ADOPT Shell's practice of enterprise-wide predictive failure analytics and computer-vision safety monitoring. We ADAPT Shell's decarbonization framework by focusing on methane reduction rather than divesting producing assets. We PILOT AI digital twin modeling across mature fields like Ozenmunaygas. However, we REJECT Shell's strategy of aggressive asset divestment and unhedged price floating, as abandoning domestic fuel subsidies would trigger severe domestic economic instability.

SECTION 5: Conclusion and Next steps

Visual Cue: Final Summary Slide with Key Takeaway and Actionable Roadmap.

KMG's immediate priority is scaling AI predictive analytics across the Atyrau and Pavlodar refineries. Success will be demonstrated by achieving a 15% reduction in unplanned outages and extending turnaround cycles by the end of 2027. Thank you.

REFERENCES (IEEE FORMAT)

[1] L. J. Gitman et al., Introduction to Business, 2nd ed. Houston, TX, USA: OpenStax, Rice University, 2024.

[2] JSC NC "KazMunayGas", Annual Report 2025: Global Standards. National Priorities, Astana, Kazakhstan, Apr. 28, 2026.

[3] Shell plc, Annual Report and Accounts 2025: Deliver More Value with Less Emissions, London, UK, Mar. 11, 2026.

[4] JSC NC "KazMunayGas", Consolidated Financial Statements for the Year Ended December 31, 2025, PwC Audit Report, Astana, Kazakhstan, Mar. 5, 2026.

[5] KMG Engineering, ABAI System Operational Brief and Digitalization Roadmap, Astana, Kazakhstan, 2026.

[6] Ministry of Energy of the Republic of Kazakhstan, Digitalization of Energy Sector and Refining Outlook, Official Report, Astana, 2026.

[7] Caspian Pipeline Consortium (CPC), Operational Crude Oil Transportation Volumes Brief, Almaty, Kazakhstan, 2026.